CLINICAL · MOUTH & THROAT

Deep neck space infection

The fascia of the neck divides it into compartments that are tidy in the anatomy book and treacherous in practice: infection from a tooth, a tonsil or a node finds the planes between them and runs, sometimes down into the chest, sometimes around the airway. Deaths from deep neck infection come from three things: a lost airway, a spreading infection that reaches the mediastinum, and a great vessel eroded or thrombosed. Secure the airway, image the neck, give the right antibiotics and drain the pus.

AT A GLANCE

- Airway plan first.
- Contrast CT, antibiotics and drainage.
- Dental disease is the usual cause.
- Watch for mediastinitis.

1 CHAPTER ONE Fascia and spaces

The map.

1.1 Deep cervical fascia

- **Superficial (investing) layer:** surrounds the neck, splitting to enclose the sternomastoid, trapezius, submandibular and parotid glands.
- **Middle (visceral) layer:** surrounds the larynx, trachea, pharynx, oesophagus and thyroid; its buccopharyngeal part covers the constrictors.
- **Deep layer:** the *prevertebral fascia* in front of the vertebrae and muscles, with the thin *alar fascia* in front of it.

All three contribute to the *carotid sheath*.

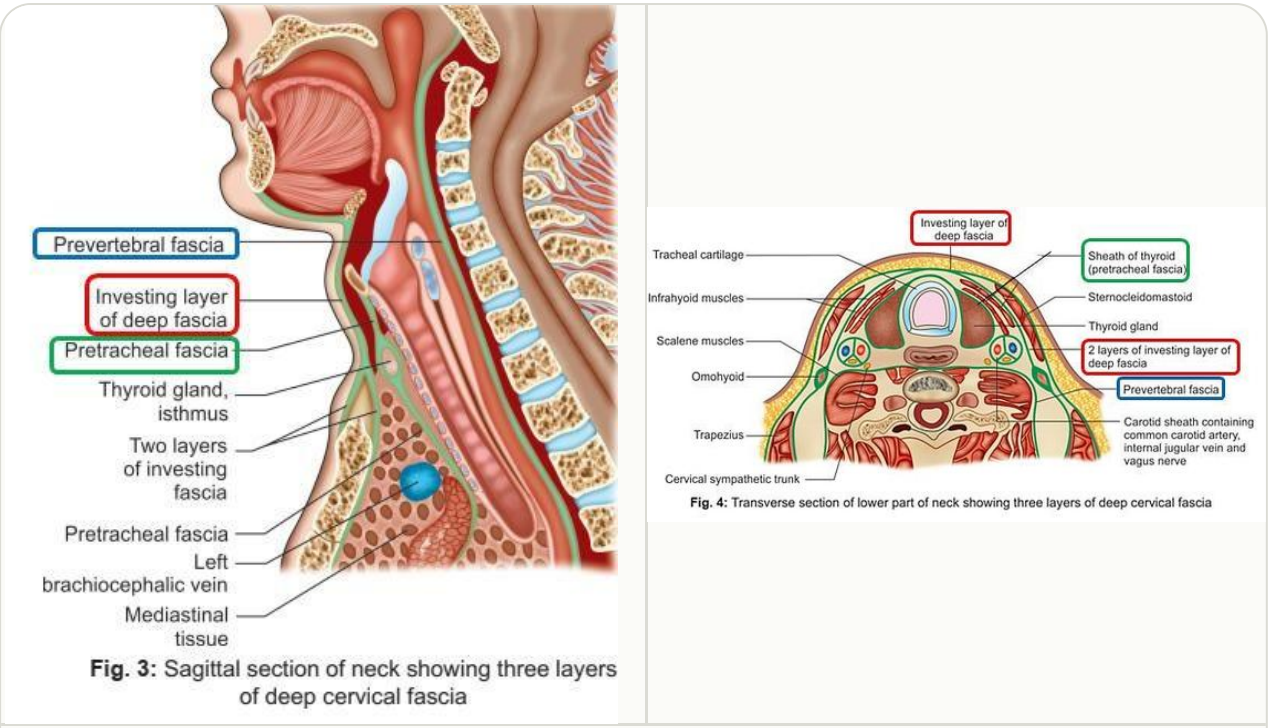


FIGURE 1 The layers of the deep cervical fascia in sagittal and transverse section.
Anatomy atlas · ENT Steps p. 274

1.2 Spaces

SPACES OF THE HEAD AND NECK

SPACE	EXTENT AND CONTENTS	USUAL SOURCE OF INFECTION
Parotid	Within the split superficial layer; gland, nodes, facial nerve, external carotid, retromandibular vein	Ascending infection via Stensen's duct in dehydrated patients
Submandibular (sublingual, submandibular, submental)	Above and below the mylohyoid, between mandible and hyoid	Teeth, sialadenitis, floor of mouth injury
Peritonsillar	Between the tonsil capsule and superior constrictor	Tonsillar crypts
Parapharyngeal	Inverted pyramid from skull base to hyoid, beside the pharynx	Tonsils, quinsy, teeth, parotid, Bezold abscess, other spaces

SPACE	EXTENT AND CONTENTS	USUAL SOURCE OF INFECTION
Retropharyngeal	Behind the buccopharyngeal fascia, in front of the alar fascia; skull base to about T2	Suppurating retropharyngeal nodes in children; trauma in adults
Danger	Between alar and prevertebral fascia; skull base to diaphragm	Spread from retropharyngeal or parapharyngeal spaces
Prevertebral	Behind the prevertebral fascia; skull base to coccyx	Tuberculosis and infection of the spine
Visceral	Around larynx, trachea, pharynx, oesophagus and thyroid, into the mediastinum	Perforation of the oesophagus, thyroiditis, surgery
Carotid sheath	Skull base to the thoracic inlet; carotid, jugular vein, vagus	Spread from adjacent spaces, drug injection
Masticator	Around the masseter, pterygoids and temporalis and the ramus	Second and third molars

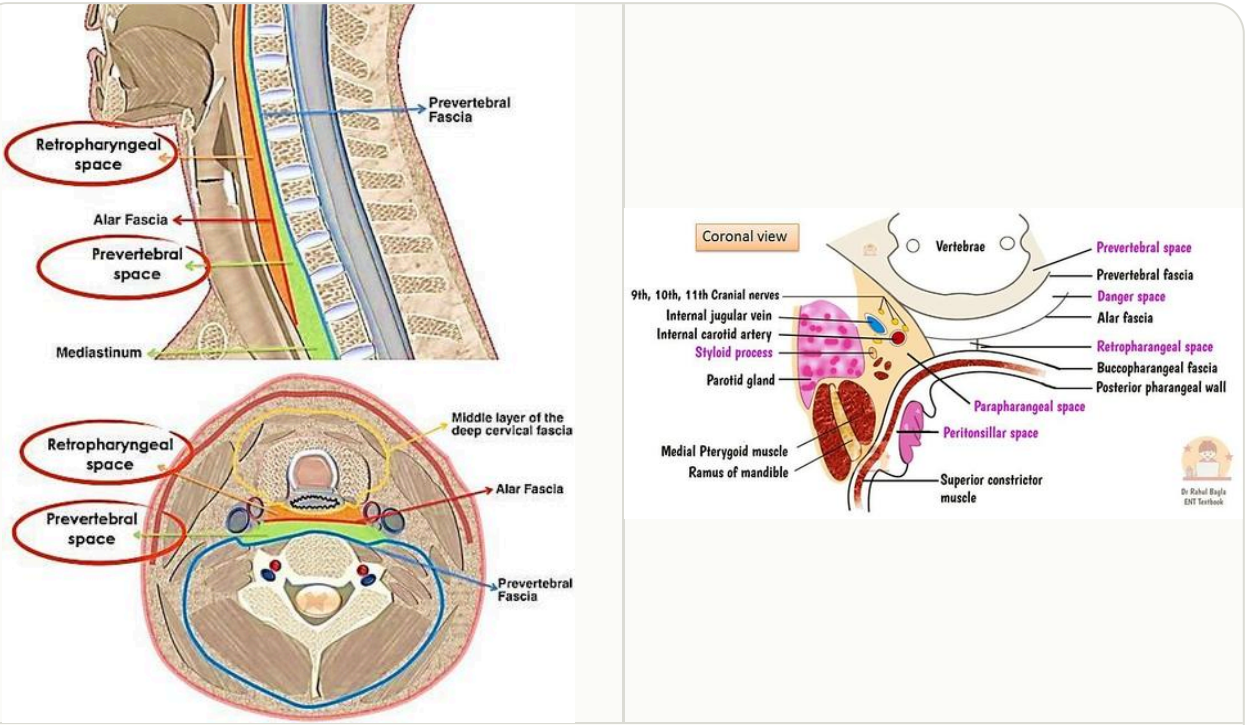


FIGURE 2 The retropharyngeal, danger and prevertebral spaces, and the parapharyngeal and retropharyngeal spaces.

Anatomy atlas · ENT Steps p. 277, ENT Steps p. 198

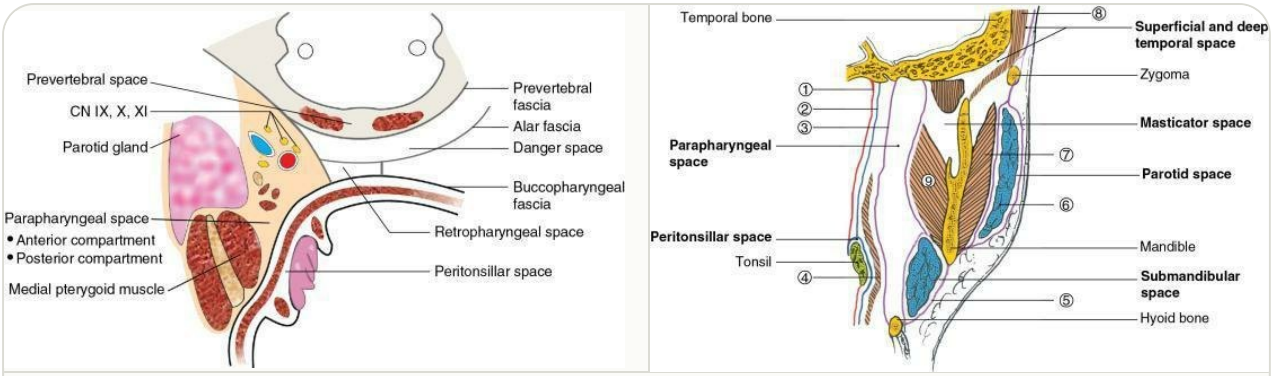


FIGURE 3 First, the spaces in relation to the pharynx where abscesses form. Second, the spaces of the head and neck in coronal section.

Dhingra, figs 52.6 and 52.9

ROUTES TO THE CHEST

The retropharyngeal space reaches the superior mediastinum; the danger space runs behind it to the diaphragm, and the carotid sheath, the "Lincoln highway of the neck", carries infection down to the root of the neck. The parapharyngeal space connects with the retropharyngeal, submandibular, parotid, carotid and visceral spaces, so it is the crossroads of spread.

CHAPTER TWO

2 General principles

Before the details.

2.1 Sources

Dental infection is the commonest source in adults, and pharyngeal and tonsillar infection in children. Others: suppurating lymph nodes, sinusitis, mastoiditis (Bezold abscess), sialadenitis, trauma, intubation and endoscopy, foreign bodies, intravenous drug injection, and infected congenital cysts. In an adult, a "neck abscess" may be a necrotic metastasis from an oropharyngeal cancer.

2.2 Organisms

Usually mixed aerobic and anaerobic mouth flora: viridans and group A streptococci, peptostreptococci, staphylococci, *Bacteroides* and *Fusobacterium*. Methicillin-

resistant *Staphylococcus aureus* is common in infants; diabetics and the immunosuppressed acquire Gram-negative organisms; tuberculosis and actinomycosis cause chronic abscesses.

2.3 Evaluation

- **History:** pain, fever, swelling, odynophagia, drooling, voice change, trismus, breathing difficulty; recent dental work, sore throat, trauma or intubation; diabetes and immune status.
- **Examination:** the teeth, floor of mouth, tonsils and posterior wall; trismus; crepitus (gas-forming organisms); cranial nerves; and *flexible endoscopy of the airway* whenever there is voice change, dyspnoea, stridor or unexplained dysphagia. Oxygen saturation can be normal with a critical airway.
- **Blood:** count, glucose, renal function, cultures.
- **Imaging:** contrast-enhanced CT is the key study, showing which spaces are involved, abscess against phlegmon, and spread to the mediastinum; clinical examination alone misjudges the spaces in many cases. Lateral neck radiographs show prevertebral swelling (over about 7 mm at C2 in adults, or more than half a vertebral body in children); chest radiographs for mediastinitis and aspiration; ultrasound in children and for guided aspiration; MRI and MR angiography for intracranial or vascular complications.

2.4 Treatment

1. **Airway first.** Sit the patient up, give humidified oxygen; steroids and nebulised adrenaline buy time. If the airway is narrowed by more than half, secure it by awake fibreoptic intubation or tracheostomy under local anaesthesia; never paralyse a patient whose airway you cannot see.
2. **Fluids:** patients are dehydrated from not swallowing.
3. **Intravenous antibiotics** at once, broad-spectrum with anaerobic cover: co-amoxiclav or ampicillin–sulbactam, or a third-generation cephalosporin with metronidazole; clindamycin in penicillin allergy and in small children; vancomycin for MRSA; antipseudomonal agents in the immunosuppressed. Adjust to cultures.
4. **Drainage** when there is an abscess in a fascial space, gas, a threatened airway, or no improvement after 48 to 72 hours of antibiotics. Small contained abscesses and phlegmon, especially in children, often settle with antibiotics alone.
5. **Remove the source:** extract the tooth, remove the foreign body.

OPERATIVE GUIDES IN THEATRE 2 operations

Drainage of deep neck abscesses and tracheostomy.

Deep Neck Abscesses
Deep neck abscess
Open Tracheostomy
Tracheostomy

3 CHAPTER THREE Ludwig's angina

The floor of the mouth on fire.

3.1 Applied anatomy

Three primary spaces surround the mandible: *sublingual* (above the mylohyoid, under the oral mucosa), *submandibular* (below the mylohyoid, under the investing fascia, containing the gland and nodes) and *submental* (between the anterior bellies of digastric). The roots of the premolars and first molar lie above the mylohyoid line and drain into the sublingual space; those of the second and third molars lie below it and drain into the submandibular space. The spaces communicate around the back of the mylohyoid and across the midline.

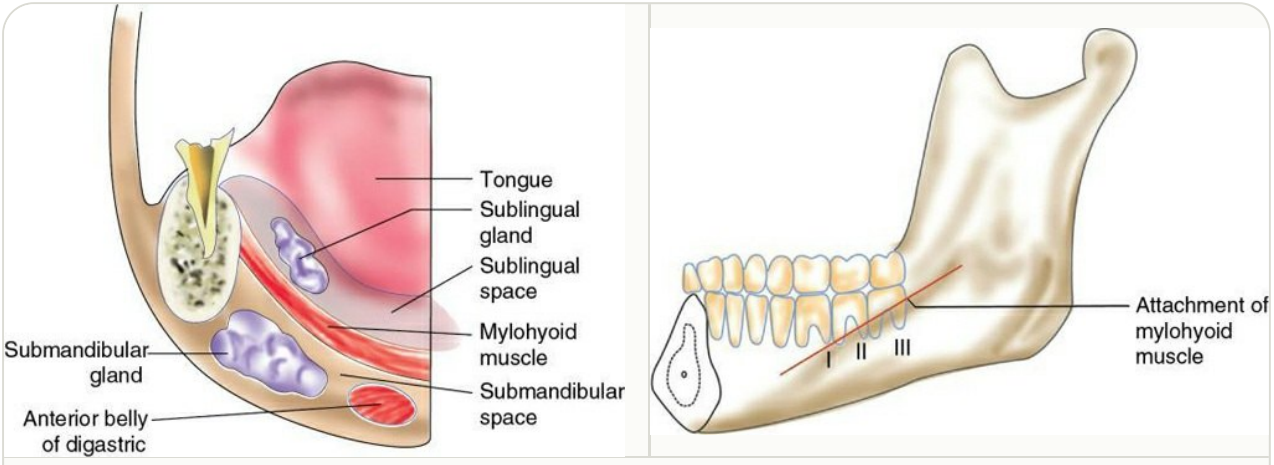


FIGURE 4 First, the submandibular space below the mylohyoid and the sublingual space above it. Second, the roots of the molars project below and those of the premolars above the attachment of the mylohyoid.
Dhingra, figs 52.1 and 52.2

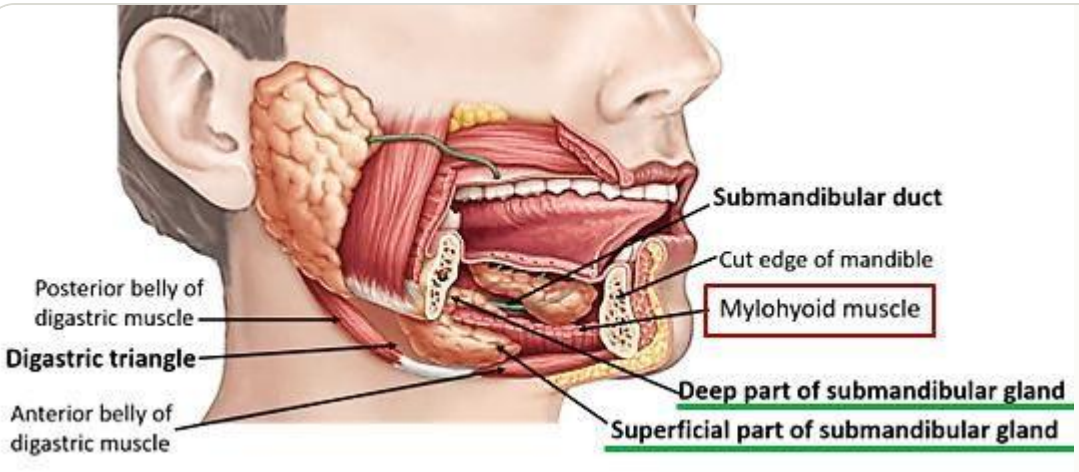


FIGURE 5 The submandibular gland around the mylohyoid.
Anatomy atlas · ENT Steps p. 189

3.2 Definition and cause

A rapidly spreading cellulitis of the submandibular, sublingual and submental spaces on both sides, usually without a discrete abscess. Dental infection accounts for about 80%; others are submandibular sialadenitis, floor-of-mouth injuries and mandibular fractures. Mixed aerobes and anaerobes.

3.3 Clinical features

Painful swallowing, drooling and trismus; a swollen floor of mouth that pushes the tongue up and back; a tense, *woody-hard*, tender swelling below the jaw on both sides; a muffled voice; the patient sits up, leaning forward. As the tongue is forced back and the larynx swells, the airway closes.



FIGURE 6 Ludwig's angina in a seven-year-old child.
Dhingra, fig. 52.3

3.4 Treatment

- Secure the airway: awake fibreoptic intubation or tracheostomy under local anaesthesia, before it is lost.
- High-dose intravenous antibiotics with anaerobic cover; steroids reduce swelling.
- Drainage: intraorally if confined to the sublingual space; externally for submandibular and submental involvement, through a transverse incision below the mandible with blunt opening of the mylohyoid. Often only serous fluid is found, yet decompression relieves the airway.
- Extract the offending tooth.

NEVER LIE DOWN A PATIENT WITH LUDWIG'S ANGINA

A sitting, drooling patient with a raised floor of mouth can obstruct completely if laid flat or sedated. Move to theatre with the airway team present.

3.5 Complications

Spread to the parapharyngeal and retropharyngeal spaces and mediastinum, airway obstruction, septicaemia and aspiration pneumonia.

CHAPTER FOUR

4 Retropharyngeal abscess

Behind the pharynx.

4.1 Applied anatomy

The retropharyngeal space lies between the buccopharyngeal fascia in front and the alar fascia behind, from the skull base to the mediastinum. A midline raphe divides it into two lateral compartments (spaces of Gillette), each with retropharyngeal nodes that receive lymph from the nose, sinuses, adenoids, nasopharynx and middle ear and that regress after about four or five years of age.

4.2 Acute retropharyngeal abscess

4.2.1 Cause

Mostly children under three to five, from suppuration of the retropharyngeal nodes after infection of the adenoids, nasopharynx, nose or sinuses. In adults, penetrating injury of the posterior pharyngeal wall or oesophagus (fish bones, instruments); rarely mastoid pus tracking under the petrous bone.

4.2.2 Clinical features

- Fever, irritability, refusal to feed, dysphagia and drooling.
- Difficulty in breathing, stridor and a croupy cough.
- *Torticollis* and neck stiffness, with the head held extended; Lee names torticollis the best indicator in children.
- A bulge on one side of the midline of the posterior pharyngeal wall.

4.2.3 Investigation

A lateral soft tissue radiograph, taken with the neck extended, shows widening of the prevertebral shadow, sometimes with gas or a fluid level. Contrast CT shows the abscess, distinguishes it from phlegmon and shows extension below the hyoid or into the mediastinum.

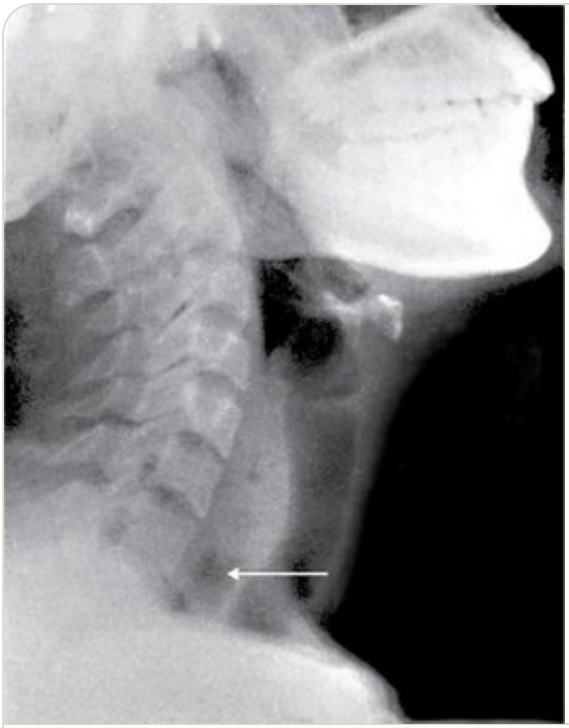


FIGURE 7 Retropharyngeal abscess: widening of the prevertebral space with gas (arrow) on a lateral radiograph.

Dhingra, fig. 52.7

4.2.4 Treatment

Intravenous antibiotics; many small abscesses and phlegmons in children resolve. A definite abscess is drained transorally with the child supine and head low, mouth held open by a gag and suction ready: aspirate first to relieve pressure, then make a vertical incision over the most fluctuant point, avoiding the carotid laterally. Under general anaesthesia, intubate gently to avoid rupture, and pack the pharynx. Tracheostomy if the airway is compromised. Low or extensive abscesses are drained through the neck.

4.3 Chronic retropharyngeal (prevertebral) abscess

Tuberculous, from caries of the cervical spine (in the midline, behind the prevertebral fascia) or from tuberculous retropharyngeal nodes (to one side, behind the buccopharyngeal fascia). Throat discomfort and mild dysphagia; a fluctuant swelling of the posterior wall; tuberculous neck nodes; radiographs show the spinal disease. Drain through the neck (along the anterior border of sternomastoid for a low abscess, the posterior border for a high one), never through the mouth when the spine is involved; antituberculous treatment, with the spinal surgeon.

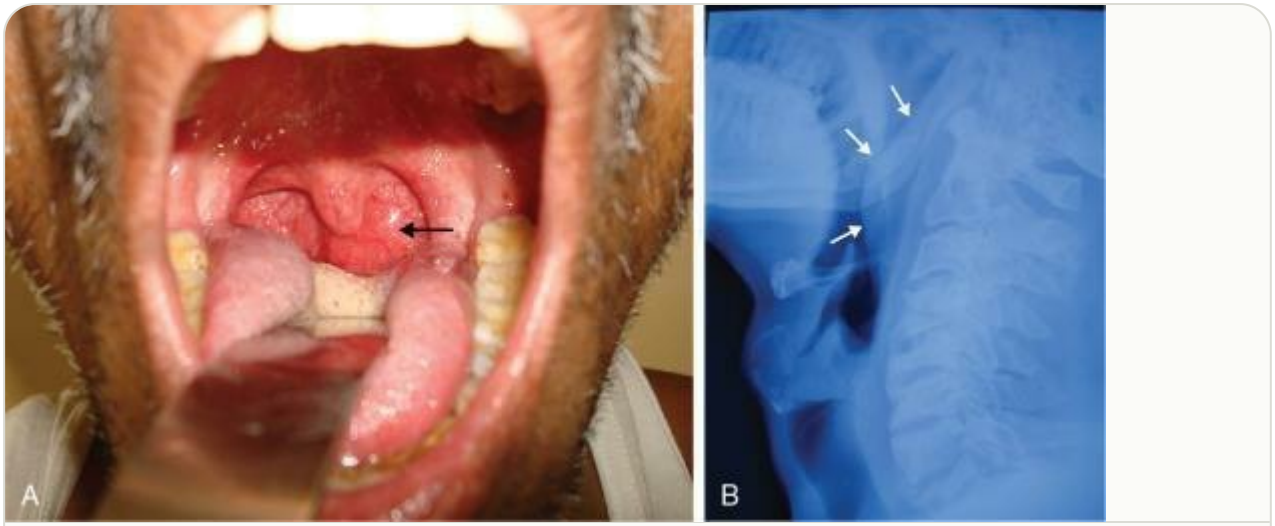


FIGURE 8 (A) A tuberculous prevertebral abscess seen in the oropharynx. (B) Radiograph of the same.

Dhingra, fig. 52.8

5 CHAPTER FIVE Parapharyngeal abscess

The lateral pharyngeal space.

5.1 Applied anatomy

An inverted pyramid with its base at the skull base and apex at the hyoid. Medially the buccopharyngeal fascia over the constrictors; laterally the medial pterygoid, ramus and deep lobe of the parotid; behind, the prevertebral fascia. The styloid process and its muscles divide it into:

- **Anterior (prestyloid) compartment:** related to the tonsillar fossa and medial pterygoid.
- **Posterior (poststyloid) compartment:** containing the internal carotid artery, internal jugular vein, cranial nerves IX to XII, the sympathetic trunk and upper deep cervical nodes.

5.2 Causes

The tonsils and adenoids, a ruptured quinsy, the last lower molar, Bezold abscess and petrositis, the parotid, retropharyngeal and submandibular spaces, and

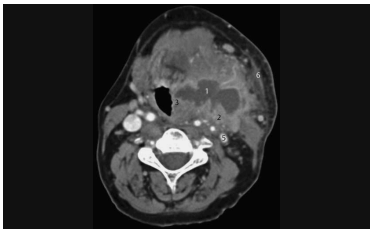
penetrating injury, including local anaesthetic injections.

5.3 Clinical features

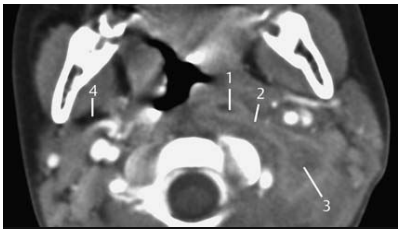
- **Anterior compartment:** the triad of medial prolapse of the tonsil and tonsillar fossa, marked *trismus* from medial pterygoid spasm, and a swelling behind the angle of the jaw; severe odynophagia.
- **Posterior compartment:** a bulge behind the posterior pillar, palsies of IX, X, XI and XII and Horner syndrome, and parotid swelling, with little trismus.
- **Both:** fever, toxaemia, sore throat and torticollis.

IMAGING IN THE APP 3 cases

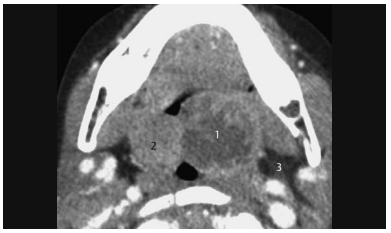
Deep neck abscesses on imaging.



Abscess of the Neck
Infection & inflammation · 1 image



Parapharyngeal Abscess
Infection & inflammation · 5 images



Tonsillar Abscess
Infection & inflammation · 7 images

5.4 Complications

Laryngeal oedema and airway obstruction; jugular thrombophlebitis and septicaemia; spread to the retropharyngeal space and down the carotid sheath to the mediastinum; mycotic aneurysm of the carotid and carotid blow-out.

5.5 Treatment

Intravenous antibiotics covering aerobes and anaerobes. Drainage under general anaesthesia (tracheostomy first if trismus is severe) through a horizontal incision 2 to 3 cm below the angle of the mandible, dissecting bluntly along the inner surface

of the medial pterygoid towards the styloid process; a drain is left. *Never drain a parapharyngeal abscess through the mouth*, for the great vessels lie in its wall.

6 CHAPTER SIX Other spaces

Parotid and masticator.

6.1 Parotid abscess

In dehydrated, postoperative or debilitated patients with poor oral hygiene, infection ascends Stensen's duct, usually *Staphylococcus aureus*. Five to seven days after an operation the parotid becomes swollen, red, indurated and tender; fluctuation is hard to feel through the thick capsule; pus can be expressed from the duct. Ultrasound or CT shows the loculi. Rehydrate, stimulate saliva, give intravenous antibiotics, and drain through a parotidectomy incision, opening the loculi bluntly in the line of the facial nerve branches. The thin deep fascia lets pus break into the parapharyngeal space.

6.2 Masticator space

Between the layers of the investing fascia around the masseter, pterygoids and temporalis, with the ramus, maxillary artery and inferior alveolar nerve; its three parts are masseteric, pterygomandibular and temporal. Infection of the second and third molars is the usual source; marked trismus is the cardinal sign. Drained through an incision lateral to the retromolar trigone, or above the zygomatic arch for the temporal space.

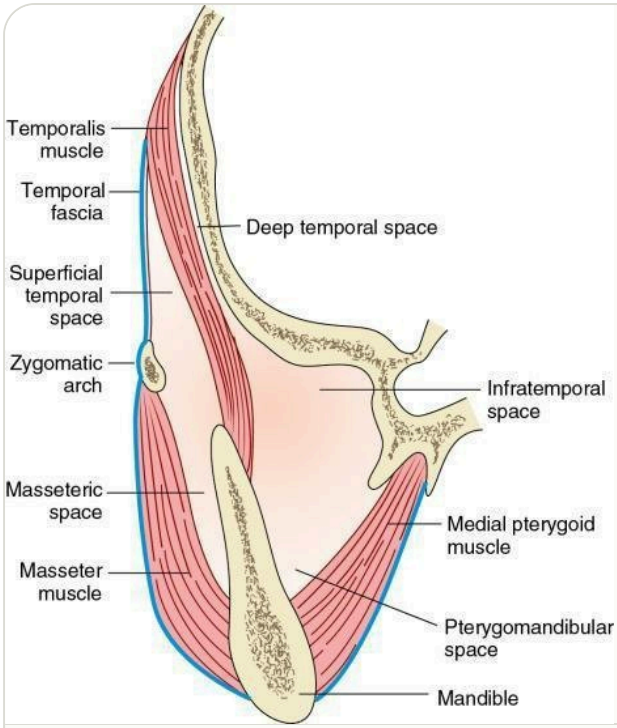


FIGURE 9 The masticator space and its three parts: masseteric, pterygomandibular and temporal.

Dhingra, fig. 52.10

7 CHAPTER SEVEN Dangerous complications

What kills.

7.1 Mediastinitis

Usually from the retropharyngeal or danger spaces: chest pain, dyspnoea, tachycardia and a widened mediastinum on the chest film. Mortality is high; combined cervical and thoracic drainage improves survival.

7.2 Lemierre syndrome

Septic thrombophlebitis of the internal jugular vein after pharyngitis, usually with *Fusobacterium necrophorum*: swinging fevers, tenderness along the sternomastoid, and septic emboli to the lungs and joints. Contrast CT shows a filling defect in the

vein. Two to three weeks of intravenous antibiotics; anticoagulation is debated; ligation if the patient deteriorates.

7.3 Carotid erosion

A mycotic pseudoaneurysm or rupture from retropharyngeal or parapharyngeal infection. Warnings: a *sentinel bleed* from the mouth, nose or ear, a pulsatile swelling, Horner syndrome or lower cranial nerve palsies. Angiography, then ligation or stenting.

7.4 Necrotising fasciitis

In the elderly, diabetic and immunosuppressed, usually dental in origin: spreading oedematous "peau d'orange" erythema with crepitus and gas on CT. Resuscitation, broad-spectrum antibiotics and repeated debridement to bleeding tissue.

7.5 Cavernous sinus thrombosis

Infection from the upper teeth or face tracking back through the valveless ophthalmic veins: fever, orbital pain, proptosis, ophthalmoplegia and a dilated pupil.

A NECK ABSCESS IN AN ADULT

After the infection settles, examine the upper aerodigestive tract. A cystic "abscess" in an adult may be a necrotic node from an HPV-related oropharyngeal cancer or an infected branchial cyst hiding one.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Layers of deep cervical fascia** and the spaces between them.
- **Ludwig's angina:** spaces, cause, airway management.
- **Retropharyngeal abscess:** acute versus chronic, radiograph, transoral versus external drainage.
- **Parapharyngeal abscess:** compartments, signs, and why it is never drained through the mouth.
- **Complications:** mediastinitis, Lemierre syndrome, carotid blow-out.

8.2 Common pitfalls

COMMON PITFALLS

- Sending a patient with a threatened airway to the CT scanner unaccompanied.
- Incising a parapharyngeal swelling through the mouth.
- Draining a tuberculous prevertebral abscess transorally.
- Calling a cystic neck mass in an adult an abscess and not looking for a primary.

Written for this app from Oxford Textbook of Otolaryngology ch. 62, Oxford Specialist Handbook ch. 12, Oxford Handbook of ENT ch. 25, Dhingra ch. 52 and K. J. Lee ch. 37. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Abscess of the Neck

IMAGING FINDINGS

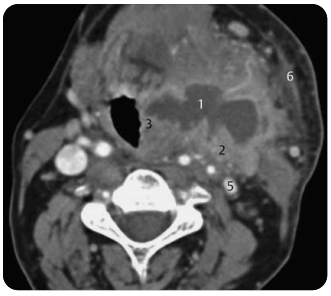
- Low-density collection with a well-formed enhancing rim
- Surrounding cellulitis and fat stranding
- The source: tooth, tonsil, node, stone or foreign body

TOP DIFFERENTIALS

- Phlegmon (no fluid centre)
- Necrotic metastatic node
- Infected congenital cyst

CHECKLIST

- Airway, mediastinum and jugular vein
- Which space: decides the drainage route



CT with intravenous contrast enhancement, axial.A large abscess (1) with a thick, contrast-enhanced capsule (2). The abscess is extendin…

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Parapharyngeal Abscess

IMAGING FINDINGS

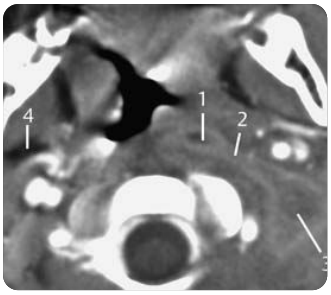
- Collection in the parapharyngeal fat, usually from the tonsil
- Medial bulge of the pharyngeal wall
- Trismus from medial pterygoid involvement

TOP DIFFERENTIALS

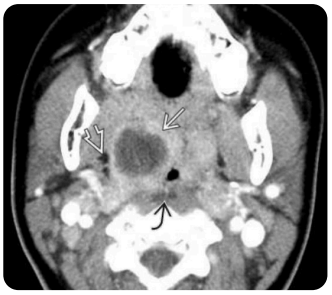
- Peritonsillar abscess
- Retropharyngeal abscess
- Deep lobe parotid infection

CHECKLIST

- Carotid space and jugular vein (Lemierre)
- Spread to the mediastinum



CT with intravenous contrast enhancement, axial. Note the bulging left pharyngeal wall, as a result of a parapharyngeal abscess (1) which is…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Tonsillar Abscess

IMAGING FINDINGS

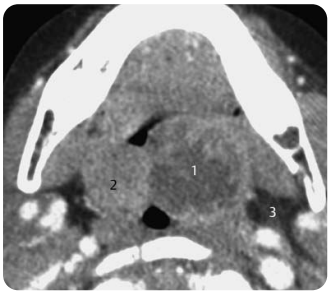
- Rim-enhancing fluid collection in or around the tonsil
- Swollen tonsil and pharyngeal wall
- Spread to parapharyngeal, masticator or carotid space

TOP DIFFERENTIALS

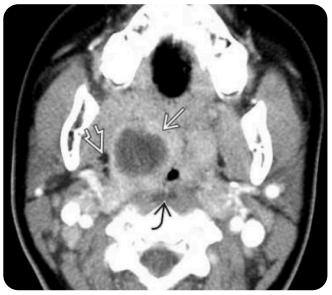
- Tonsillitis without abscess (phlegmon)
- Tonsillar retention cyst
- Retropharyngeal abscess; tonsillar carcinoma

CHECKLIST

- Airway
- Internal jugular vein thrombosis and carotid



CT with intravenous contrast enhancement, axial. The left tonsil (1) is much larger than the contralateral side (2) with central hypodensities…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.